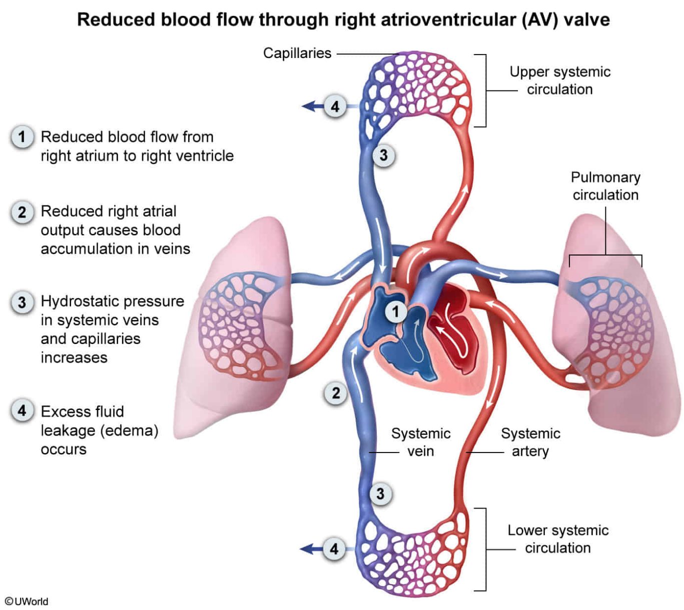
A patient is found to have reduced blood flow through pulmonary vessels and excess fluid accumulation in the abdomen and legs. Which of the following conditions would best explain this patient's symptoms?

- A. Reduced blood flow through the heart valve separating the right atrium and right ventricle [39%]
- B. Vasodilation of arteries supplying blood to abdominal organs [16%]
- C. Increased volume of blood pumped from the right ventricle [4%]
- D. Backflow of blood through damaged valves in systemic arteries [39%]

Correct
39% answered correctly

Explanation:



The heart consists of four chambers (two atria and two ventricles) and continuously contracts to pump blood through the circulatory system. The left atrium receives oxygenated blood returning from the lungs and transfers this blood into the left ventricle, a highly muscular chamber that pumps blood throughout the body via systemic arteries. Deoxygenated blood returns to the heart via systemic veins and fills the right atrium, which transfers blood into the right ventricle. Right ventricular contraction pumps deoxygenated blood to the lungs via the pulmonary arteries.

Specialized valves facilitate **unidirectional** blood flow through the heart. **Atrioventricular (AV) valves** allow blood to flow from the atria to the ventricles, and **semilunar valves** allow blood to flow from the ventricles into the arteries.

In this scenario, a patient exhibits reduced pulmonary blood flow and excess fluid accumulation in the abdomen and legs. These symptoms are best explained by **reduced blood flow through** the **heart valve separating** the **right atrium and right ventricle** (ie, right AV valve). This condition would *decrease* the volume of blood loaded into the right ventricle prior to contraction, *reducing* blood flow from the right ventricle into the pulmonary arteries.

The reduced volume of blood flowing into the right ventricle and through the pulmonary circuit would in turn cause a backup of blood in the right atrium and systemic vessels. Accordingly, blood volume and hydrostatic pressure in systemic veins would *increase*, causing *excess* fluid leakage from systemic vessels into the surrounding interstitial fluid and subsequent fluid accumulation (edema) around tissues.

(Choice B) Vasodilation of arteries supplying blood to abdominal organs occurs under normal circumstances (eg, after a meal) and does *not* by itself impair pulmonary blood flow or lead to swelling.

(Choice C) An increased volume of blood pumped from the right ventricle leads to increased (*not* decreased) blood flow through pulmonary vessels.

(Choice D) Systemic veins (*not* arteries) have valves that, when damaged, can lead to backflow (ie, flow away from the heart) of blood.

Educational objective:
Deoxygenated blood returning to the heart fills the right atrium, which transfers blood to the right ventricle. The right ventricle pumps blood to the lungs via the pulmonary circuit.

Time spent:0 sec	**QID**:402599
Subject: Biology	**System:** Circulation and Respiration